A Concentric Loop Convolutional Neural Network for Manual Delineation-Level Building Boundary Segmentation From Remote-Sensing Images

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***Abstract*— To date, accurate building footprint delineation in the surveying, mapping, and geographic information system (GIS) communities has been dependent on human labor. In this article, to address this issue, we propose a concentric loop convolutional neural network (CLP-CNN) method for the automatic segmentation of building boundaries from remote-sensing images. The proposed method consists of three components: 1) a boundary detector to extract coarse polygonal boundaries of individual regions of interest; 2) a concentric loop-shaped convolutional network with bidirectional pairing loss to fine-tune the vertices of the polygons; and 3) a refinement block, which removes redundant vertices and regularizes the boundaries to polygons at the manual delineation level. We also demonstrate that the proposed CLP-CNN method is applicable to other generic objects in natural images. Experiments on two building datasets confirmed that more than 77%/67% of the building polygons predicted by the proposed method are on par with the manual delineation level, representing a significant saving in the labor cost of manual annotation. In generic object boundary delineation tests performed on the Semantic Boundaries Dataset (SBD), the proposed method outperformed the most recent state-of-the-art methods by at least 3.1% in average precision (AP). Furthermore, compared with other vertex matching methods, the learning process of the proposed method converges faster. The source code will be available at http://gpcv.whu.edu.cn/data.**

***Index Terms*— Building boundary extraction, convolutional neural network, instance segmentation, polygon regularization.**

I. Introduction

AUTOMATIC building boundary extraction from remote-sensing images, which can benefit geographic information system (GIS) map production and updating, urban planning, and population estimation, has been studied for more than 30 years [1]. However, mature methods are still lacking, due to the imperfect imaging conditions, as a result of the atmospheric dynamics, seasonal changes, building structure diversities, and background complexity. To date, the building footprints used in GIS systems have been manually delineated.

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Deep learning is the fastest-growing technology in remote-sensing imagery analysis. Recent studies indicate that the feature representations learned by deep learning methods are highly effective in object detection [3], [4], map production and updating [5]–[8], and image denoising [9]. Deep learning-based semantic segmentation methods for automatic building extraction have been seen as holding great promise. The mainstream semantic segmentation methods [10]–[14] predict a binary building segmentation map by the use of a fully convolutional network (FCN) [15]. A few of these methods have a post-processing step that involves extracting and vectorizing each foreground region of the segmentation map as individual building polygons [16], [17]. However, dense pixel-wise classification requires a huge amount of calculation, and a key drawback of these FCN-based methods is that the extracted building boundaries are often zigzagged, broken, and far from the manual delineation level.

Directly extracting the edges and corners of individual buildings is an alternative approach for pixel-wise segmentation, which has been popularized by the emerging contour-based approaches for boundary extraction or instance segmentation. The early studies attempted to extract corners as key points to form the topology of an object of interest [18], [19], but this approach is not suitable for complex buildings. The PolarMask method [20] introduces a polar coordinate centered at the object center, and predicts the boundary of the object with dense distance regression based on the limited features at the object center, resulting in the loss of contour details. Moreover, the polar coordinate form with a low degree of freedom also reduces the upper bound of the performance. LSNet [21] follows the idea of PolarMask and adjusts the contour vertices from the features of the object center. The state-of-the-art graph convolutional network (GCN) [22], [23] methods, such as Curve GCN [24], [25], or the special circular convolution (CIR-Conv) network methods such as Deep Snake [2] and DANCE [26], can adjust the boundaries of the object in an iterative manner. By providing a set of vertices for a coarse boundary and the corresponding image, these methods involve extracting the high-level features of each vertex, to input them into a designed learnable module, and to output the offset.

However, these GCN-based or circular convolution-based methods are still in an early stage and have obvious shortcomings. First of all, these methods only utilize limited information, i.e., the features of the boundary vertices [the yellow

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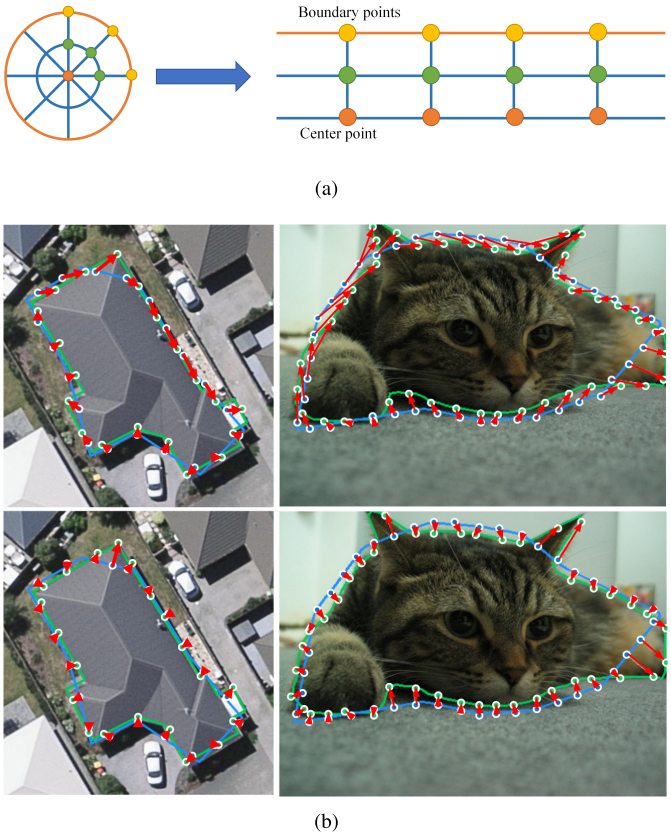


Fig. 1. Two novel contributions of this article. (a) Concentric loop convolution structure, where the whole unfolded concentric loop is processed, whereas the existing methods only process the boundary (the yellow dots in the orange line). (b) Pairing method for the predicted and ground-truth vertices, where the top row is the result of a competing method [2] and the bottom row is the result of the proposed method.

vertices in Fig. 1(a)], for vertex adjustment, and ignore the information of the other parts, e.g., the inside of the object, which limits their learning ability. Secondly, these methods employ coarse and even unreasonable boundary supervision strategies. In fact, the proper supervision of boundary adjustment (through the loss functions of the discrete boundary vertices) is the key to iterative learning. However, unlike pixel-level segmentation, it is difficult to design a differentiable calculation process to measure the distance between the predicted and ground-truth boundaries. In these studies [2], [24], [26], a vertex is designated as the starting point, and each predicted vertex is then paired to the ground-truth vertex, in a clockwise or anti-clockwise sequence. However, this simple approach fixes the pairs of ground truth and predicted vertices in advance, regardless of the dynamic changes of the predicted vertices, and creates many improper correspondences [see Fig. 1(b) as an example], which greatly increases the learning difficulty. In this article, to handle these two problems, we attempt to make full use of the information provided in the imagery to boost the network performance, and effectively supervise the predicted boundaries with a novel pairing strategy to reduce the learning difficulty.

In this article, we propose a novel deep learning-based method for automatic building boundary extraction. The method consists of three stages: 1) a coarse boundary extraction stage; 2) a concentric loop structure stage for polygon adjustment; and 3) a refinement stage. In this method, a novel concentric loop structure is proposed to fully utilize the boundary and inner information of the object of interest, which differs from the previous contour-based methods that only utilize the information of the center points [20], [21] or boundaries [2], [24]–[26]. As shown in Fig. 1(a), by discretizing the rays originating from the center with vertices, the set of vertices consists of a concentric loop. The concentric loop structure aggregates the global inner and boundary information of an object to achieve high-precision boundary regression. To handle the concentric loop, we unfold the loop into a regular grid, which is then processed by the 2-D concentric loop convolutional neural network (CLP-CNN) to adjust the positions of the boundary vertices. The difference between the 1-D circular convolution structure [2] [only the orange boundary in Fig. 1(a)] and the proposed concentric loop structure is analogous to that between the single circular chromosome in bacteria and archaea and the many chromosomes in eukaryotes.

For the possible mismatch of the predicted and ground-truth vertices, we propose a new bidirectional method to pair the optimal corresponding points. In short, the ground-truth vertices are sequentially paired with the nearest unpaired predicted vertices, while at the same time addressing the "one-to-many" matching problem that is caused by automatic pairing. The distance between each pair then constitutes the loss. This method ensures that correct matching [Fig. 1(b)] and rapid convergence are achieved.

In addition to handling the two key problems in the current contour-based deep learning approaches, the third part of our work focuses on the properties of buildings, i.e., the man-made well-structured boundaries. To ensure flexible boundary adjustment for the concentric loop network, the redundant vertices are densely interpolated at the initial boundaries. However, most of the vertices are unnecessary, and can lead to many small waves and zigzags. We propose two things to deal with this problem. Firstly, we use a CIR-Conv network [2] to distinguish whether a vertex is valid or not, and to keep the corner points only. Secondly, the corners and edges are adjusted outside the learning loop, to reach a manual delineation level, according to an empirical polygon regularization method [16].

The main contributions of this work are summarized as follows.

We propose an end-to-end contour-based method named the concentric loop convolutional neural network (CLP-CNN) method for the automatic segmentation of building boundaries. The concentric loop structure is newly designed for extracting more complete information of a building instance, instead of utilizing only the features of the center points or boundaries, as in previous works.

A new bidirectional method is proposed to pair the predicted and ground-truth vertices, which reduces the learning difficulty. We also introduce a refinement block, which translates the boundaries of buildings to more concise polygons by removing redundant vertices.

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The proposed CLP-CNN method demonstrated a state-of-the-art performance in the experiments conducted in this study. The results confirmed that more than 77% (on the WHU building dataset) and 67% (on the Inria building dataset) of the building polygons predicted by the proposed method are on par with the manual delineation level. Moreover, the proposed method not only showed a superior performance in the remote-sensing building datasets, but it also surpassed the state-of-the-art methods in the Semantic Boundaries Dataset (SBD) [27] made up of natural images.

II. Related Work

In this section, we briefly review the development of building extraction methods, the progress in the contour-based instance segmentation methods, and the vertex matching strategies in boundary loss calculation, as well as the inherent problems of the existing methods.

Most of the current FCN-based building extraction methods [5], [10]–[14] still focus on binary segmentation. However, in our work, we are concerned with both the instance segmentation performance and how close the extracted polygons are to manually delineated ones. A few researchers [17], [28]–[30] have studied the instance segmentation of buildings, but they all made use of only Mask R-CNN [31]. Few studies have focused on structured building polygon extraction. In the work of Wei *et al.* [16], FCN-based segmentation was used for generating a binary segmentation map, and each connected foreground region of the map was vectorized and regularized to form individual building polygons. However, limited by the capabilities of the FCN and the high density of buildings, the connected regions are prone to be irregular or broken, which affects the extraction of complete and accurate building boundaries. Golovanov *et al.* [33], Iglovikov *et al.* [34] added auxiliary branches to better transform the semantic segmentation results into instance segmentation results, but complex post-processing is still needed.

The contour-based segmentation methods are attracting increased attention, although, to date, they have not been applied to structured building extraction. Compared to pixel-based segmentation, the contour-based methods have the advantages of fewer parameters and less post-processing. In earlier studies [34], [35], the authors tried to use contour-based methods to obtain instance segmentation results, but the performance could not be compared with that of the pixel-based segmentation methods. Several studies [18], [19], [36] all used a recurrent neural network (RNN) [37] to predict the locations of the key points of an object, which is an approach that often suffers from the loss of some vertices. Ling *et al.* [24] introduced a GCN into the boundary point adjustment, and obtained a performance that was equivalent to that of the pixel-based methods. However, this method is not a fully automatic instance segmentation method, and it does not make full use of the advantages of the graph map structure. Peng *et al.* [2] and Liu *et al.* [26] made improvements on the basis of [24] by replacing the original GCN structure with a CIR-Conv structure. In addition to these two-stage methods, contour-based single-stage methods such as PolarMask [20], DARNet [38], and the methods proposed in [40] and [41] have also achieved good results. However, all of these one-stage and two-stage methods use the features of the vertices as the input, which is an approach that does not fully utilize the available information because the inner area of an object also provides useful information. The concentric loop convolution (CLP-Conv) structure proposed in this article makes full use of all the information that can be accessed, to achieve better boundary extraction results.

A key to the task of contour-based segmentation extraction is how to calculate the loss of a boundary effectively and efficiently, i.e., how to pair the corresponding predicted and ground-truth vertices. In the pixel-based methods, the difference between the predicted result and the ground truth is directly measured by comparing the coordinate values of the corresponding pixels. However, it is difficult to design a differentiable calculation process to measure the overall difference between contours. The polar coordinate-based method [20] allows the direct calculation of the overall contour loss through an approximate intersection over union (IoU) calculation, but this is invalid for discrete contour points with high degrees of freedom. The contour-based methods often use an ordered set of points to represent the edges, and the quality of the predicted result can thus be evaluated by calculating the coordinate difference between the predicted and label vertices. Therefore, correct vertex pairing is critical for calculating the correct distance between the predicted and label boundaries, the efficiency of the contour adjustment, and the convergence ability. However, the vertex pairing approach has not been deeply explored. Ling *et al.* [24] simply paired the vertices by minimizing the global coordinate difference between the predicted and ground-truth vertices in a sequential manner; however, this strategy can lead to many unreasonable vertex pairs. Peng *et al.* [2] designated the first ground-truth vertex and paired it with the nearest predicted vertex, and then matched the rest of the vertices in sequence. Although the calculation process is simplified, this method has the same mismatch problem as the method proposed in [2]. Wei *et al.* [40] proposed three matching methods, namely, the nearest point, nearest line, and corner point with projection methods. Compared with the methods proposed in [2] and [24], these three strategies can eliminate many unreasonable pairs, but they are prone to ignoring points in the valleys or hills. In the proposed method, unlike these unidirectional vertex pairing strategies, we propose a bidirectional method considering both the overall boundary and local details, which not only obtains optimal pairing results and a finer boundary representation, but also greatly speeds up the training process.

III. Method

In this section, we describe the three main parts of the proposed building boundary extraction method, namely, the coarse boundary extraction, the concentric loop network for vertex adjustment, and the vertex reduction and boundary regularization. The workflow of the proposed CLP-CNN method is shown in Fig. 2. Examples of the building extraction process are shown in Fig. 3.

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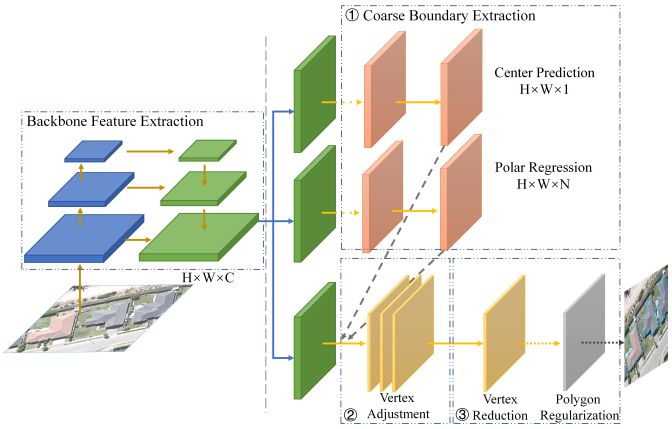


Fig. 2. Overview of the CLP-CNN method.

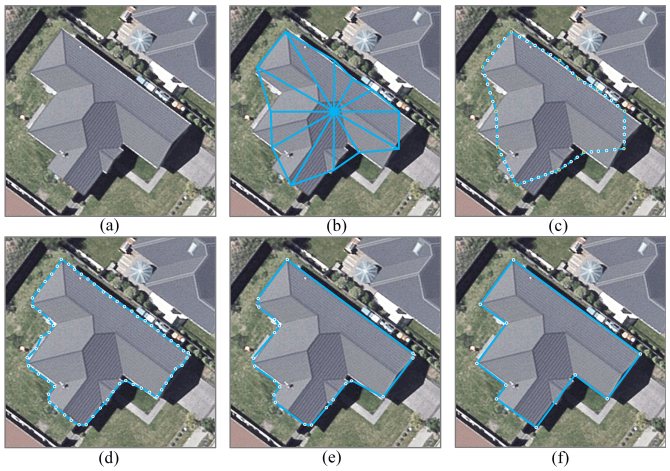


Fig. 3. Examples of the building extraction process. (a) Input image. (b) Polar polygon predict. (c) Contour interpolation. (d) Vertex adjustment. (e) Vertex reduction. (f) Polygon regularization.

A. Coarse Boundary Extraction

The high-level convolutional features are first extracted by a backbone network. A feature pyramid is generated (the left part of Fig. 2), the parameters of which are the same as those in [2]. We then adopt the PolarMask method [20] to extract the initial boundaries. The last features of the backbone network are used to predict the center point and a set of boundary vertices. As shown in Fig. 3(b), the center point and the boundary points are extracted, and the boundary points are then densified [Fig. 3(c)]. Please refer to [20] for more details. The polar coordinates of the boundary points are converted to Cartesian coordinates for the subsequent processing.

B. Concentric Loop Convolution for Vertex Adjustment

PolarMask [20] and LSNet [21] regress the coordinates of the boundary vertices of an object based on the features at the center point, whereas Deep Snake [2] and DANCE [26] only regress the coordinates based on the boundary features. In contrast, we propose the CLP-CNN method, which can make full use of the boundary and inner feature information for vertex adjustment. The structure of the concentric loop is shown in Fig. 1(a) and the CLP-CNN method is depicted in Fig. 4. According to the initial contour and center point

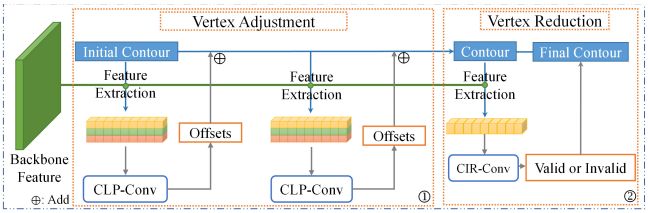


Fig. 4. Structure of the proposed CLP-CNN method.

(the envelope center of the current contour), an s-band ($s = 3$) equidistant concentric loop is generated based on the center point and the corresponding contour points, and then unfolded to a grid, as shown in Fig. 1(a). The coordinates of the concentric loop points and center point are recalculated when the contour is updated. We then equidistantly interpolate N points from the initial boundary represented by the M equiangular vertices ($M = 12$) in each loop, and the features of the center point repeat N times. In practice, we counted the number of building corner points in many building datasets and found that more than 95% of the buildings had less than 20 corner points. Therefore, we choose $N = 60$ to accommodate buildings with more vertices and ensure the flexibility of the boundary adjustment. We also tested 128 vertices, as used by Deep Snake, and the result was similar to that when using 60.

In fact, the unfolding of a loop which is represented by N vertex features results in periodic discrete signals, similar to the readings on a clock. The 1-D feature of each vertex extracted from the last features of the backbone is denoted as the signal f, and the unfolded periodic feature can be defined as

$$(f_N)_i \triangleq (f_{i+w})_N, \quad w \in Z \tag{1}$$

where N is the number of vertices and Z is the set of integers, as used in [2].

We can then extend the processing of a loop to a concentric loop. The unfolding result of a concentric loop is a periodic grid. The second dimension is the band number of the concentric loop s.

$$(f_N)_{i,j} \triangleq (f_{i+w,j})_N, \quad w \in Z, \ 0 < j \leq s. \tag{2}$$

Using a convolution kernel as a learning unit, we have

$$(f_N * k)_{i,j} = \sum_{w=-\alpha}^{\alpha} \sum_{h=-\beta}^{\beta} f_{(i+w,j+h)} k_{w,h} \tag{3}$$

where k: $[(-\alpha, \alpha), (-\beta, \beta)]$ is the 2-D learnable convolutional kernel. In our work, both the features (the colored 3-band blocks in Fig. 4) and the convolution are expressed in 2-D space, instead of one dimension. The CLP-Conv structure is then introduced to process the 2-D features, i.e., all the available information of an object region, to predict the offset of each vertex.

In the CLP-Conv structure, we use standard 2-D convolutional layers to learn the adjustment of the vertices. The only difference is that it is necessary to copy and concatenate the feature of the starting vertex to the end of the last vertex, and vice versa, creating a periodic structure for the convolution operation. The exact structures and parameters of

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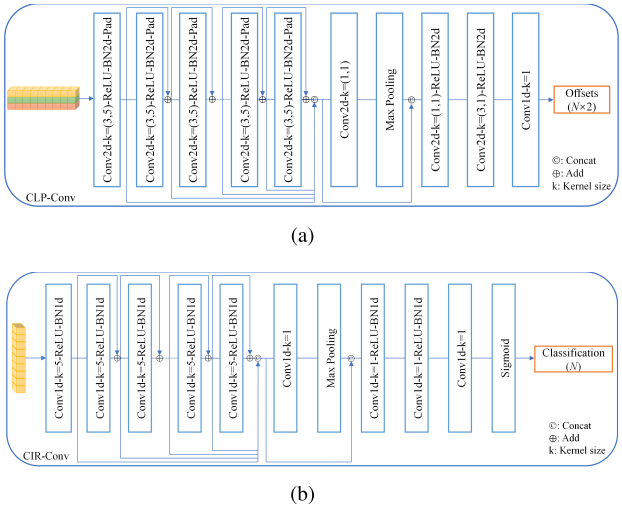


Fig. 5. Structures and parameters of the vertex adjustment module. (a) Layers and parameters of the CLP-Conv structure for vertex adjustment. (b) Layers and parameters of the CIR-Conv structure for vertex reduction.

the CLP-Conv structure are shown in Fig. 5(a). The output of the CLP-Conv structure is a coordinate offset, which is then added to the initial boundary. The CLP-Conv operation is applied twice to refine the offsets of the vertices in a stepwise manner. It should also be noted that it was experimentally proved that more stages are not necessary.

C. Vertex Reduction and Polygon Regularization

We interpolate dense points in the boundary to ensure flexible boundary adjustment, but this creates many redundant vertices at the same time. To handle this, we introduce a learnable vertex reduction module. After the vertex adjustment is executed twice, the features of the adjusted boundary (yellow blocks) are input into a 1-D CIR-Conv network, to judge whether each vertex is valid or not. We regard a vertex as valid if its distance to a real building corner is less than P pixels; for example, $P = 3$. The network structure is shown in Fig. 4(b), which is similar to the one used in [2] for vertex adjustment. The boundary is then reduced by excluding the invalid vertices, to obtain the final output of the network. It should be noted that vertex reduction is a simple binary classification problem, i.e., judging whether a vertex is a building corner or not. We chose CIR-Conv instead of CLP-Conv to simplify the calculation.

There is still a gap between the output boundary of the CLP-CNN method and a manually delineated one, in that the former may have redundant vertices near the corners and slightly biased vertices [e.g., Fig. 3(e)]. We apply a simple but effective main direction-based polygon regularization method [16] for the fine adjustment of the vertices and edges of a building. The edges of a building are classified into two kinds according to a given threshold (such as 6 m) for the edge length, i.e., long edges and short edges. A main direction list is created and initialized with the direction of the longest edge, and a new direction for a long edge is added to the list if the angle between it and any of the directions in the list is larger than a given threshold. All of these edges are then rotated around their centers to the parallel or perpendicular directions of their closest main directions. Please refer to [16] for more details. Finally, we obtain the structured building polygons, as shown in Fig. 3(f).

D. Bidirectional Matching Loss

It is demonstrated in Fig. 1(b) that the single-directional pairing of predicted and ground-truth vertices creates improper correspondences, which can result in difficulties for the vertex adjustment and network convergence. We therefore introduce a bidirectional vertex matching algorithm to predict the optimal pairs of predicted and ground-truth vertices. The corresponding loss is called the bidirectional matching loss.

We first sort through the ground-truth vertices and find their nearest matches in the predicted vertex set. However, it is possible that many of the ground-truth vertices will correspond to only one predicted vertex. For such a case, we only keep the pair with the maximum distance. This approach can endow the ground-truth vertices on a contour valley or hill with correct matches [see the left ear of the cat in Fig. 1(b)]. We then sort through the predicted vertices, and pair those vertices that have not been matched with the nearest ground-truth vertices. This strategy guarantees that all of the predicted vertices x^p, including the corners, are paired to the most reasonable ground-truth vertices x^{gt}. The set of matched vertices $[x^p, x^{gt}]$ represents the whole boundary. We use the l_1 distance to describe the loss of the boundary, as shown in (4). The loss is calculated after every adjustment of the vertices, to release the potential of each CLP-Conv module and speed up the convergence. In contrast, in [2] and [24], the loss was calculated only once at the last vertex adjustment

$$L_{\text{bid}} = \frac{1}{N}\sum_{i=1}^{N} l_1\left(x_i^p - x_i^{gt}\right). \tag{4}$$

The other losses in the network are set as follows. The loss of the center point prediction in the coarse boundary extraction remains the same as that of CenterNet [41], i.e., the focal loss, which is denoted as L_c. For the polar regression, we use both smooth l_1 loss [42] and polar IoU loss [20] with a ratio of 0.3:1. For the vertex reduction, we use the focal loss, denoted as L_p, to reduce the impact of the imbalance between the positive and negative samples.

IV. Experiments

A. Datasets and Implementations

We used the WHU aerial building dataset [16] and the Inria aerial dataset [43] for the building boundary extraction testing. The WHU dataset features a large quantity of highly accurate building labels. The dataset covers an area of 450 km^2 in Christchurch, New Zealand, and includes more than 187 000 buildings of various architectures and purposes. All of the aerial orthogonal images have been seamlessly cropped into 1024 × 1024 tiles at a 0.2-m ground resolution. The training set, validation set, and test set contain 2793 (130 000 buildings), 627 (14 500 buildings), and 2220 (42 000 buildings) patches, respectively. The Inria aerial dataset contains different structures and styles of buildings. Subdatasets of two regions, the Tyrol in Austria and Kitsap County in the US, were selected from the Inria dataset for the testing. For each

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region, there are 36 5000 × 5000 aerial images with a 0.3-m ground resolution. We divided the images of each region into training/test sets at a ratio of 3:1.

We also used the SBD dataset [27] to demonstrate the generalizability of the proposed method.

In this article, we use the standard average precision (AP) index to evaluate the boundary extraction (instance segmentation) accuracy. AP_{50} and AP_{75} are also listed. However, how to judge whether an extracted building boundary approaches the manual delineation level is another question. We assume a 2-pixel or 3-pixel manual delineation error introduced for each edge, both of which create an instance-level IoU threshold for each building. This assumption is technically reasonable as manual delineation rarely achieves a subpixel accuracy, and is further verified via the visual investigation of the predicted results.

All the experiments were executed on a desktop computer with an Nvidia TITAN RTX 24 GB graphics processing unit (GPU). The network implementation was based on PyTorch. DLA-34 [44] was used as the backbone, and the hyperparameters were set the same as those of CenterNet [41]. The proposed network was trained with enhanced images (i.e., multi-scale data augmentation, flipping, and cropping, as used in Deep Snake) with a size of 512 × 512 pixels, and tested on the original images with a size of 1024 × 1024 pixels. The learning rate was initially set to $1e^{-4}$ and decreased by 0.5 at 90 and 120 epochs. We trained the coarse boundary extraction network for 40 epochs, and then trained the whole network for 110 epochs.

B. Comparison With the State-of-the-Art Methods

We compared the proposed CLP-CNN method with the most recent boundary extraction and instance segmentation methods on the WHU and Inria datasets. It should be clarified that, in these experiments, we only tested the performance in general instance segmentation. The refinement step, including the vertex reduction and regularization, was not included, as these processes were designed only for reaching a manual delineation level for structured buildings. The results are listed in Table I, where it can be seen that the performance of the proposed CLP-CNN method exceeds that of all the other methods, and it outperforms the second-best method (deep snake [2]) by 3.4% in AP on the WHU dataset and 3.9% in AP on the Inria dataset. Compared with Deep Snake, which suffers from limited information inputs and a poor vertex pairing strategy, the proposed CLP-CNN method benefits from a more reasonable contour initialization module, the CLP-CNN structure that makes full use of the boundary and inner information of objects, and a superior bidirectional ground-truth/predicted vertex pairing strategy. In the results listed in Table I, the earlier Mask R-CNN method [31] and the semi-automatic Curve GCN method present a poor performance. Curve GCN [24] requires manually input bounding boxes (in the experiments, Mask R-CNN was used to provide the bounding boxes of the buildings), and its performance is unstable. PolarMask [20] with the DLA-34 backbone extracts relatively coarse boundaries and lacks a fine-tuning module, which limits its performance.

TABLE I

PERFORMANCE OF THE DIFFERENT METHODS ON THE WHU AND INRIA DATASETS

Method	WHU dataset			Inria dataset		
	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
Mask R-CNN [32]	58.1	77.4	67.9	41.3	72.3	43.9
Curve GCN* [24]	58.3	76.5	68.6	37.3	58.6	41.7
PolarMask [20]	63.7	84.6	76.2	49.5	80.2	57.8
Deep Snake [2]	67.7	85.0	78.3	55.9	81.3	64.7
CLP-CNN	71.1	85.3	79.7	59.8	81.5	67.9

* indicates that the bounding boxes required by the semi-automatic Curve GCN method were provided by Mask R-CNN.

TABLE II

ABLATION STUDY WITH THE WHU DATASET

Method	AP	AP_{50}	AP_{75}
Baseline	67.1	85.1	78.2
+ Polar initial	68.6	84.9	79.0
+ CLP-Conv	70.1	84.5	79.6
+ Bidirectional matching loss	70.9	85.3	79.6
+ Multi-stage loss	71.1	85.3	79.7

We also calculated the efficiency of the CLP-CNN method. On the Nvidia TITAN 24 GB GPU, the CLP-CNN method can process 10 1024 × 1024 test images per second (i.e., 10.0 fps), which is a fast speed. Deep Snake with a simpler structure obtains 14.9 fps. However, in a typical offline remote-sensing image segmentation, the accuracy is more critical.

C. Ablation Study

An ablation study was conducted on the WHU dataset, using the Deep Snake method with two-stage CIR-Conv, instead of the three-stage convolution originally used as the baseline, which caused a 0.6% decrease in AP compared to the original Deep Snake method (see Table I). The number of vertices was set to 60. It should also be noted that more stages of CLP-Conv do not improve the accuracy of the final model.

1) Initial Boundaries: The initialization of the boundaries has a great impact on the further adjustment of the vertices. After replacing the initialization method of Deep Snake with the 12-ray PolarMask method [20], the AP is improved by 1.5% (Table II). We also tested the use of eight rays, for which the improvement is 1.4%. The second column of Fig. 6 shows the initial boundaries generated by CLP-CNN and Deep Snake. Compared with the octagon contours used in Deep Snake, the initial boundaries obtained by CLP-CNN with a higher degree of freedom are closer to the boundaries of the buildings, and can cope with more complex building structures.

2) Concentric Loop Convolution Versus Circular Convolution: When replacing the circular convolution (CIR-Conv) structure in Deep Snake with the concentric loop convolution (CLP-Conv) structure, the AP score is incrementally improved by 1.5% (a 3.0% improvement from the baseline), as shown in Table II. This demonstrates that the inner area of an object can provide effective information for better boundary extraction. The internal information of an object can also help to define the current instance, so as to avoid incorrect vertex adjustment across buildings in dense areas. The third and fourth columns of Fig. 6 show the boundaries after the first and second vertex adjustments, where the boundaries adjusted by CLP-Conv are closer to the ground truth than those adjusted by CIR-Conv.

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Fig. 6. Intermediate results of CLP-CNN (blue) and Deep Snake (red). First column: ground-truth polygons (yellow); Second column: initial boundaries; Third column: boundaries after the first vertex adjustment; Fourth column: boundaries after the second vertex adjustment.

3) Number of Concentric Loops: We tested different numbers of concentric loops to discover the optimal concentric loop structure. According to Fig. 7, compared with using a single loop (the boundary loop, $s = 1$), the AP score is significantly improved by adding the center-point loop ($s = 2$). When $s = 3$ (as used in the experiments) or 4, the AP reaches a maximum. Using more loops results in a slight decrease in AP, which indicates that excessive interpolation can increase the learning difficulty. Therefore, we used three loops in this study.

4) Vertex Pairing Strategy: Replacing the single-directional pairing rule used in [2] and [24] with the proposed bidirectional vertex pairing method results in a 0.8% improvement in AP. Please note that this is the incremental result of having applied the proposed boundary initialization and CLP-Conv on the baseline. In fact, if we only change the vertex pairing rule on the baseline, the AP improvement reaches 1.5%, which demonstrates that a reasonable vertex pairing strategy improves the quality of the extracted boundaries. Moreover, by providing correct adjustment directions and optimal descent routes [see Fig. 1(b)], the training becomes more robust and easier to converge. We discovered that, around the 60th epoch, the accuracy curve of the training reaches its peak. In contrast, the accuracy curve of training based on the single-directional pairing rule requires more than 100 epochs.

5) Multistage Loss Versus Single Loss: Setting the losses for all of the CLP-Conv modules, instead of just the last one, as in other works [2], [24], contributes to a 0.2% improvement in AP. Although the improvement is not significant, supervising each CLP-Conv module is very convenient.

6) Vertex Reduction: Adding the vertex reduction module leads to a 0.6% decrease in AP. However, the redundant points are greatly reduced, and only 30% of the vertices remain. The precision loss can be attributed to the loss of tiny details. However, to approach the manual delineation level of well-structured buildings, vertex reduction is necessary. In previous studies, the classic nonlearning-based Douglas–Peucker algorithm [45], which requires empirical parameter setting, was commonly used [16]. We show in Section IV-D that the learning-based vertex reduction strategy functions slightly better than the Douglas–Peucker algorithm, without requiring additional parameter setting.

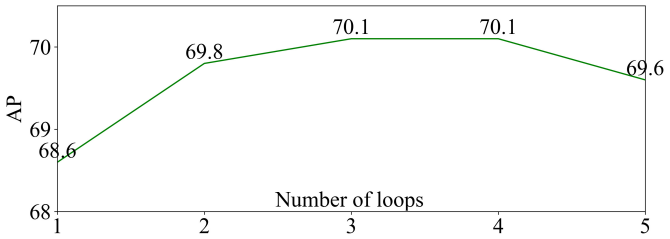


Fig. 7. Evaluations on different numbers of loop in the 2-D CLP structure.

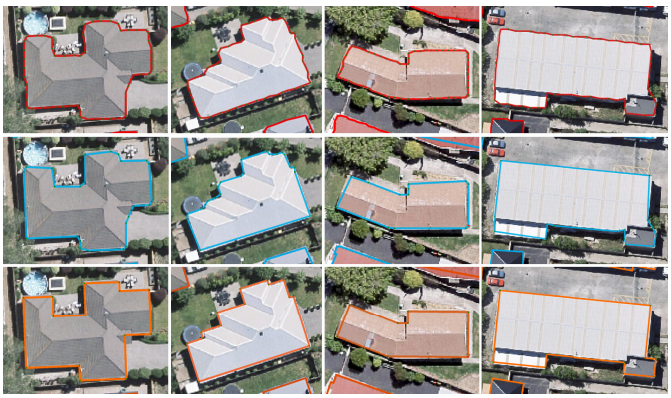


Fig. 8. Examples of the final extraction results. First row: boundary extraction results; Second row: vertex reduction results; Third row: polygon regularization results (the final results of the proposed algorithm). The third building is an example of a nonrectangular polygon.

D. Manual Delineation-Level Building Boundary Delineation

After adding the refinement step, including vertex reduction and polygon regularization, we can finally evaluate how many extracted buildings reach the manual delineation level. In Table III, we can see that 77.8%/67.7% of the buildings can be delineated as reaching the manual delineation level at a two-pixel delineation accuracy on the WHU and Inria building datasets, respectively. If the restriction is loosened to three pixels, 85.2%/75.7% of the buildings reach the level of manual delineation. It can be observed that implementation of the learning-based vertex reduction strategy slightly outperforms the use of the conventional Douglas–Peucker algorithm. Furthermore, using the learning-based vertex reduction strategy, the empirical trial-and-error approach used to discover the optimal parameters for the Douglas–Peucker algorithm can be avoided. The performance difference between the two datasets is due to the fact that the Inria dataset has a lower imaging quality, more shading trees, and worse discriminability between buildings and backgrounds.

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Fig. 9. Extracted polygons in a large area of the WHU dataset (a) and Kitsap (b) and Tyrol (c) in the Inria dataset. Red polygons: not reached manual level; yellow polygons: reached 3-pixel accuracy; blue polygons: reached 2-pixel accuracy.

Fig. 8 shows the effects of the boundary extraction, vertex reduction, and polygon regularization, where it can be observed that the building samples with both simple and complex shapes are accurately extracted.

Fig. 9 is a large map showing the final extraction results, where the red polygons are those that have not reached the manual delineation level. The yellow polygons are those which have reached a 3-pixel accuracy, and the blue polygons are those which have reached a 2-pixel accuracy. It can be observed that the predicted yellow and blue polygons have no visual difference with the manually delineated ones, and even many of the red polygons seem to be delineated satisfactorily. The visual inspection confirms the rigor and correctness of the threshold setting for the manual delineation level. Although the architectural styles of the three regions are different, most of the buildings can be accurately extracted, indicating the prospect of replacing manual work with an automated algorithm. There are, however, certain factors that caused a few cases of failure in the Kitsap and Tyrol images, i.e., buildings of a very small size tend to be neglected and the parts shaded by trees are difficult to delineate accurately.

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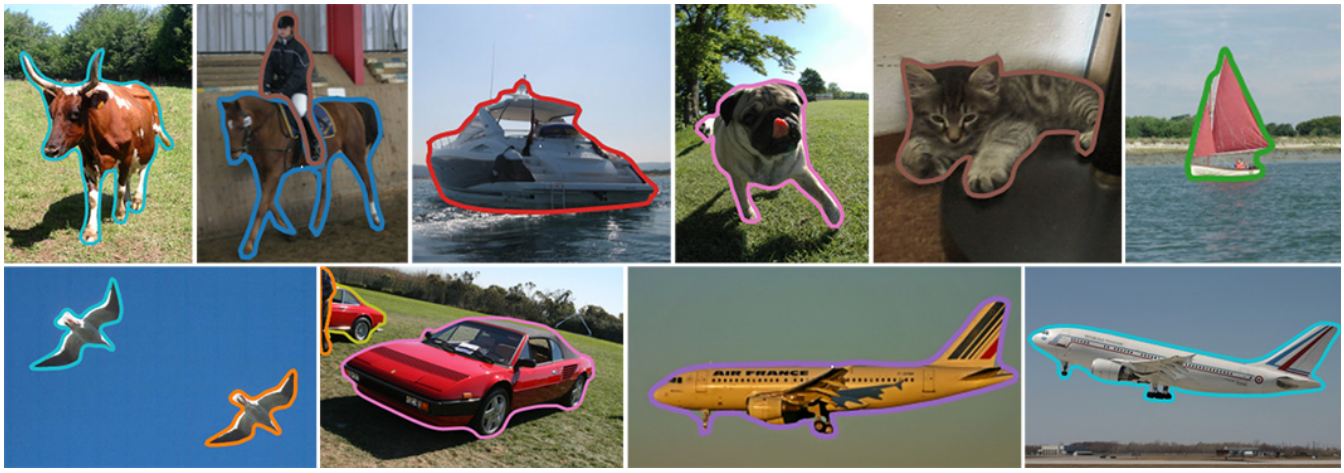


Fig. 10. Examples of the boundary extraction results obtained by the proposed CLP-CNN method with the SBD dataset.



(a) (b) (c) (d) (e)

Fig. 11. Examples of buildings that are not correctly extracted by the proposed method. (a) Building with an opening in the roof. (b) Undetected building. (c) Basketball court mistaken as a building. (d) and (e) shaded buildings.

TABLE III

RATIO OF EXTRACTED BUILDINGS THAT REACH THE MANUAL DELINEATION LEVEL

Method	WHU dataset		Inria dataset	
	2 pixels	3 pixels	2 pixels	3 pixels
Proposed	77.8%	85.2%	67.7%	76.7%
Proposed*	77.4%	85.2%	67.0%	76.6%

* indicates that the learning-based vertex reduction was replaced with the Douglas-Peucker algorithm (with the parameter tolerance empirically set to 1.5 times of the image spatial resolution).

TABLE IV

RESULTS OBTAINED WITH THE SBD DATASET

Method	AP	AP_{50}	AP_{75}	Backbone
ESE [36]	35.3	40.7	12.1	Darknet-tiny
FCIS [47]	-	65.7	52.1	ResNet101-C5
PolarMask [20]	47.0	58.1	30.4	DLA34
Deep Snake [2]	54.4	62.1	48.3	DLA34
CLP-CNN	57.5	66.2	52.2	DLA34

PolarMask, Deep Snake, and the proposed CLP-CNN method share the same encoding backbone (DLA34).

E. Experiments With the SBD Dataset

The proposed CLP-CNN method is not only suitable for building extraction, but can also be used as a general method for instance segmentation. To verify this, we compared the proposed method with other recent methods on the SBD dataset, which is made up of natural images. The results are listed in Table IV. It is noteworthy that we removed the vertex reduction module and the polygon regularization operation, which are not necessary in a natural image dataset. To ensure that the comparison was conducted on the same baseline, the light feature extraction backbone (DLA-34) was uniformly applied in PolarMask, Deep Snake, and the proposed CLP-CNN method.

The accuracy of the proposed CLP-CNN method exceeds that of the other methods, with the proposed method having a 3.1-point lead in AP over the second-best-performing Deep Snake method. The results confirm the superiority of the proposed method on this natural image dataset. Examples of the boundary extraction results obtained by the proposed CLP-CNN method on the SBD dataset are shown in Fig. 10.

The other important thing which should be noted is that we only trained the proposed model for 100 epochs, but Deep Snake was trained for 160 epochs. This further demonstrates the effectiveness of the bidirectional vertex pairing method with regard to rapid convergence.

F. Limitations

Although the excellent performance of the proposed CLP-CNN method has been demonstrated, it is unavoidably impacted by certain factors, including unpredictable backgrounds, indistinguishable buildings and backgrounds, occlusions, etc. Fig. 11 shows a few examples where the proposed method failed. In Fig. 11(a), it can be seen that the proposed method cannot accurately extract the building with an opening

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in the roof, as the goal of the proposed method is to extract the outer contour. For such a rare case, a special strategy should be considered. Fig. 11(b) is an example of a missed detection case, where the center-point detector [41] fails. Although designing a more powerful center-point detector is not the focus in this article, we will attempt to fix this problem as the detection of center points is a prerequisite of the proposed method. Fig. 11(c) is a false detection case, where a basketball court is mistaken as a building. To solve this problem, we should create more training samples, especially those negative samples that can tell the model the subtle difference between buildings and similar backgrounds. The proposed method is robust to partial occlusion, as shown in Fig. 11(e); however, it fails in Fig. 11(d) due to the poor image quality and the indistinguishable building and background. In fact, most of the building is shaded by trees and surrounded with trees, which is extremely challenging, even for manual delineation.

V. Conclusion

In this article, we have presented a concentric loop convolutional network (CLP-CNN) method for structured building boundary extraction from remote-sensing images. The concentric loop structure provides comprehensive information about an object for the accurate localization of the boundaries, whereas, in previous studies, only the boundary features have been used for vertex adjustment. The proposed bidirectional vertex pairing method discovers the best matches between the ground-truth vertices and the predicted vertices of the boundaries, which greatly speeds up the convergence. The refinement module consisting of vertex reduction and polygon regularization operations adjusts the extracted boundaries to man-made well-structured shapes. In the experiments conducted in this study, the proposed CLP-CNN method not only predicted more than 77% and 67% of the building polygons in the WHU and Inrial datasets at the manual delineation level, which indicates that the proposed method represents a bright prospect for replacing this current labor-intensive task, but the proposed method also showed its superiority over the state-of-the-art methods in a natural image dataset.

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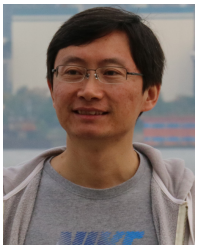
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